virtusa Jatayu

Jatayu Season 6

NIYAM

Agentic Regulatory Circular to Live Rule Engine

Team Name: **Dragorithm**

Team Members:

1.Senaaravichandran A
2.Abiyshek S
3.Jerrick Austin Manuel G
4.Venkat Narayanan S
5.Vishal R

Mentor Details

Prof. Suresh R

College

Sri Manakula Vinayagar Engineering College,
Madagadipet, Puducherry, India.

Problem Statement: FDS NA, Use Case 2

Title of the Use Case	Description	Details
Use Case 2: Regulatory Circular to Live Rule Engine	**Problem Statement:** Banks must manually interpret regulatory circulars, identify applicable obligations, map them to internal policies and existing system rules, and coordinate with IT to implement changes. This process can take weeks or months, risks missing critical obligations, and may introduce incorrect rule scope or unexpected transaction impacts. **Description:** A multi-agent AI system reads regulatory circulars, determines applicability, extracts obligations, maps them to internal rules, drafts changes, backtests their impact, and prepares them for human approval. Approved changes are deployed through shadow mode, controlled rollout, monitoring, and rollback.	**Why Agentic AI:** Specialized agents collaborate via A2A to interpret regulations, map policies, create rules, analyze impact, and iteratively resolve issues. **Workflow:** Circular Intake → Obligation Extraction → Policy Mapping → Rule Authoring → Backtesting → Impact Analysis → Validation → Human Approval → Shadow Deployment → Rollout & Monitoring. **Data Sources:** Regulatory circulars, internal policies, rule repositories, bank profiles, transaction data, backtest results, deployment metrics, and approval history. **Tech Stack:** Multi-agent orchestration, document retrieval, obligation registry, rule DSL, A2A messaging, sandbox engine, backtesting, version control, deployment APIs, monitoring, and LLMs. **Guardrails:** Draft-only rule changes, clause-level traceability, validation, mandatory human approval, controlled rollout, monitoring, rollback, and rate limits. **Enterprise Impact:** Faster regulatory implementation, fewer compliance gaps, reduced manual effort, better traceability, and safer rule deployment.

Our Answer:
NIYAM

- **7 A2A Agents** — a Conductor and six workers, found via Agent Cards
- **MCP Tool Layer** — per agent allowlists, no direct system access
- **Knowledge Graph** — clause to rule traceability with a Bank Semantic Model
- **Safe Release** — two person approval, shadow, canary, auto rollback

NIYAM: Agentic Regulatory Circular to Live Rule Engine

Problem Statement

Banks interpret every RBI circular by hand, map it to policies and system rules, then wait on IT to change them. This takes weeks to months, risks missed obligations and wrong rule scope, and causes unexpected transaction impact with no clause level audit trail.

Business Need/Opportunity

- **Circular to Rule Automation:** agents read circulars, extract obligations and draft rule changes in hours instead of months
- **Zero Missed Obligations:** every MUST clause maps to a rule or a signed manual control, with clause level traceability
- **Safe Change Deployment:** backtesting, impact analysis, shadow mode and canary rollout with auto rollback before customers are affected
- **Inspection Ready Compliance:** every live alert traces back to the circular sentence and the humans who approved it

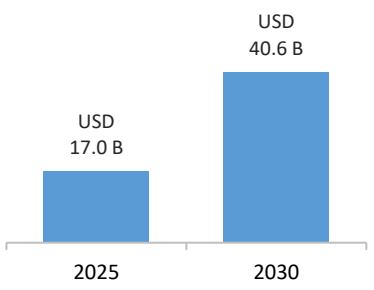
Target Segment

- **Urban Cooperative & Small Finance Banks:** full RBI regulatory load with lean compliance teams
- **NBFCs (Mid and Upper Layer):** fast changing scale based rules and digital lending norms
- **Large Banks:** a regulation to rule layer over existing AML, fraud and limit engines
- **BFSI System Integrators:** an accelerator for compliance transformation programs

Market Size and Growth

Global RegTech Market

Market size in USD Billion, analyst midpoint, CAGR ~19%

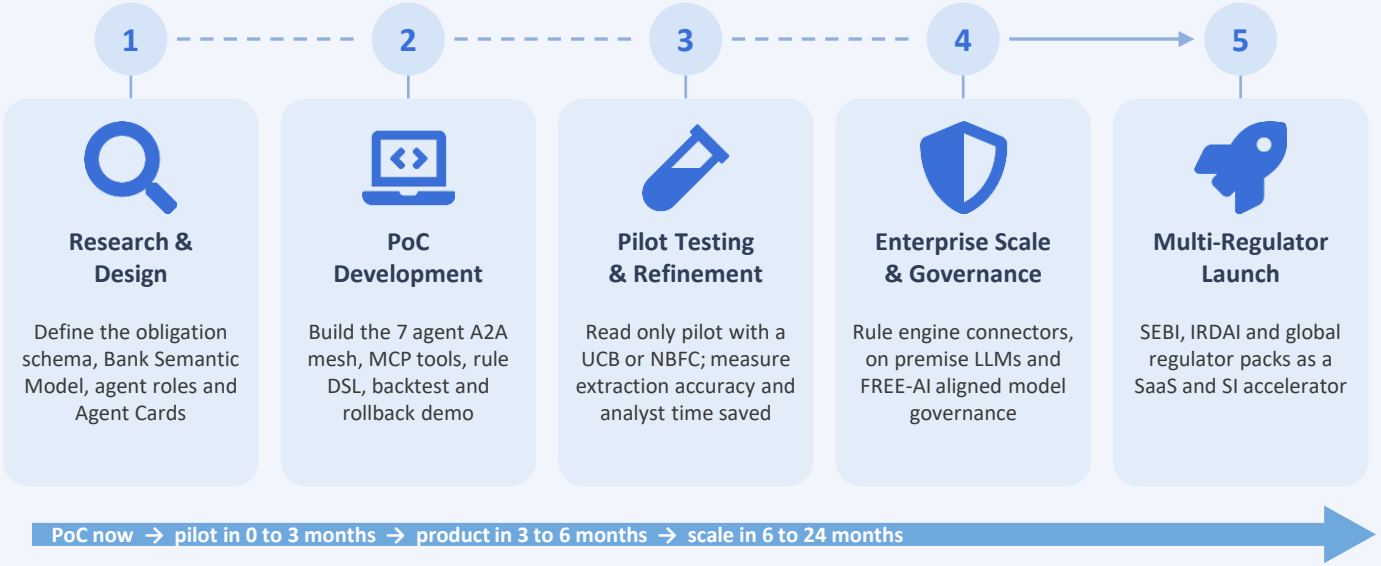


Sources: IMARC, ResearchAndMarkets (2025); 2030 implied at ~19% CAGR

Market Overview

Study Period	2025 to 2030
Global Market (2025)	USD 15 to 19 Billion
Global CAGR	17% to 21%
India RegTech CAGR	18% to 27% (2025 to 2030)
RBI Master Directions	244, from 9,000+ circulars
RBI Penalties (FY25)	353, worth ₹54.78 cr
Target Entities	1,457 UCBs, 9,000+ NBFCs
India SAM (est.)	₹550 cr per year
Key Players	CUBE, Corlytics, IBM

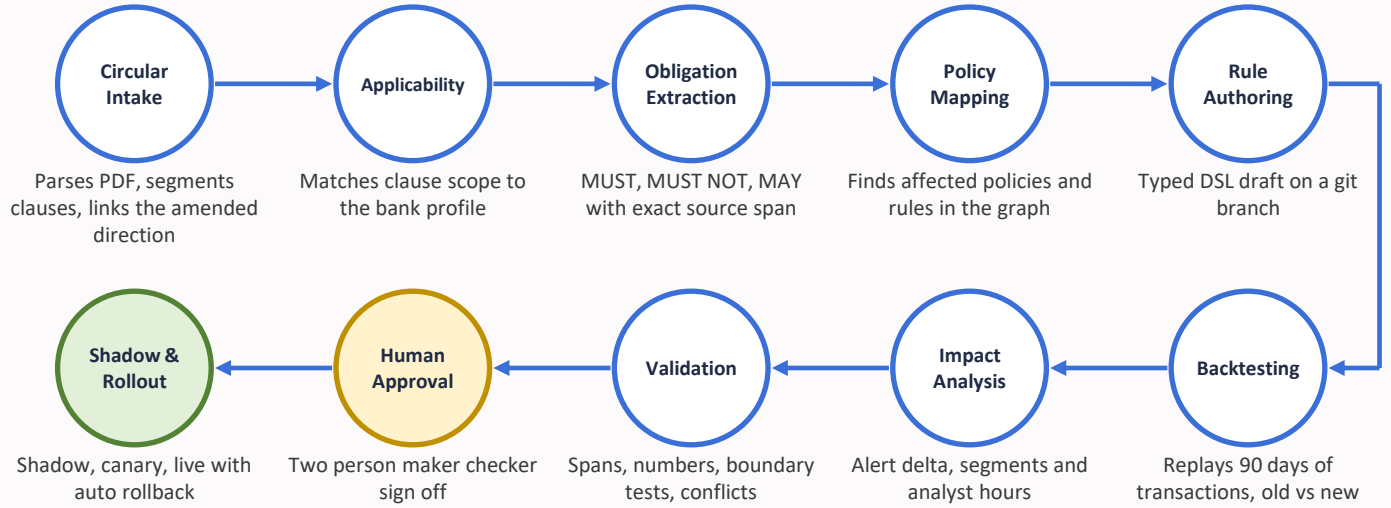
Future Readiness



Solution Details

- **Multi-Agent Regulatory Intelligence:** seven agents (a Conductor and six workers) read, scope and extract obligations from each circular and collaborate over the A2A protocol, discovered through their Agent Cards.
- **Knowledge Graph & Semantic Model:** a Regulatory Knowledge Graph, modelled for Neo4j, links circulars, clauses, obligations, policies and rules, while a Bank Semantic Model maps regulatory language to the bank's own data.
- **Safe Rule Lifecycle:** rules are drafted in a typed DSL, backtested on transaction history, validated, approved by two humans, then released through shadow, canary and live with auto rollback.
- **Governed by Design:** tools are reached only through MCP with per agent allowlists; every action is logged with clause level provenance under token and iteration limits.

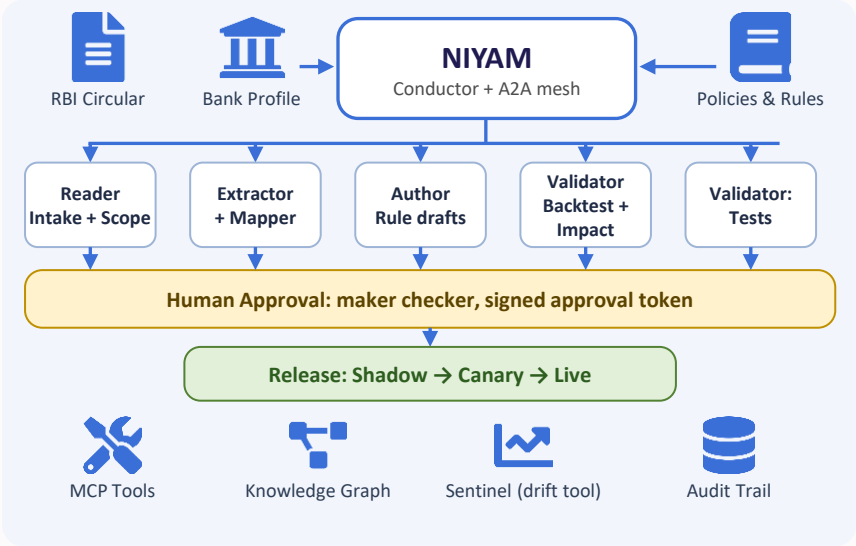
Circular to Rule Compilation Engine (CRCE)



Unique Selling Proposition

1. **Circular to Live Rule:** covers the full path from PDF to a deployed rule, not just an obligation register.
2. **Tests From the Regulation:** auto generated boundary tests catch threshold, comparator, unit and date errors.
3. **Human in Control:** two person approval; agents can never deploy on their own.
4. **Golden Thread Audit:** any alert traces to its clause, rule version, backtest and approvers.
5. **Plug and Play Regulators:** new regulators join as Agent Cards and MCP servers, with no rewrite.

Working



Use Case Diagram

